A Tractable State-Space Model for Symmetric Positive-Definite Matrices

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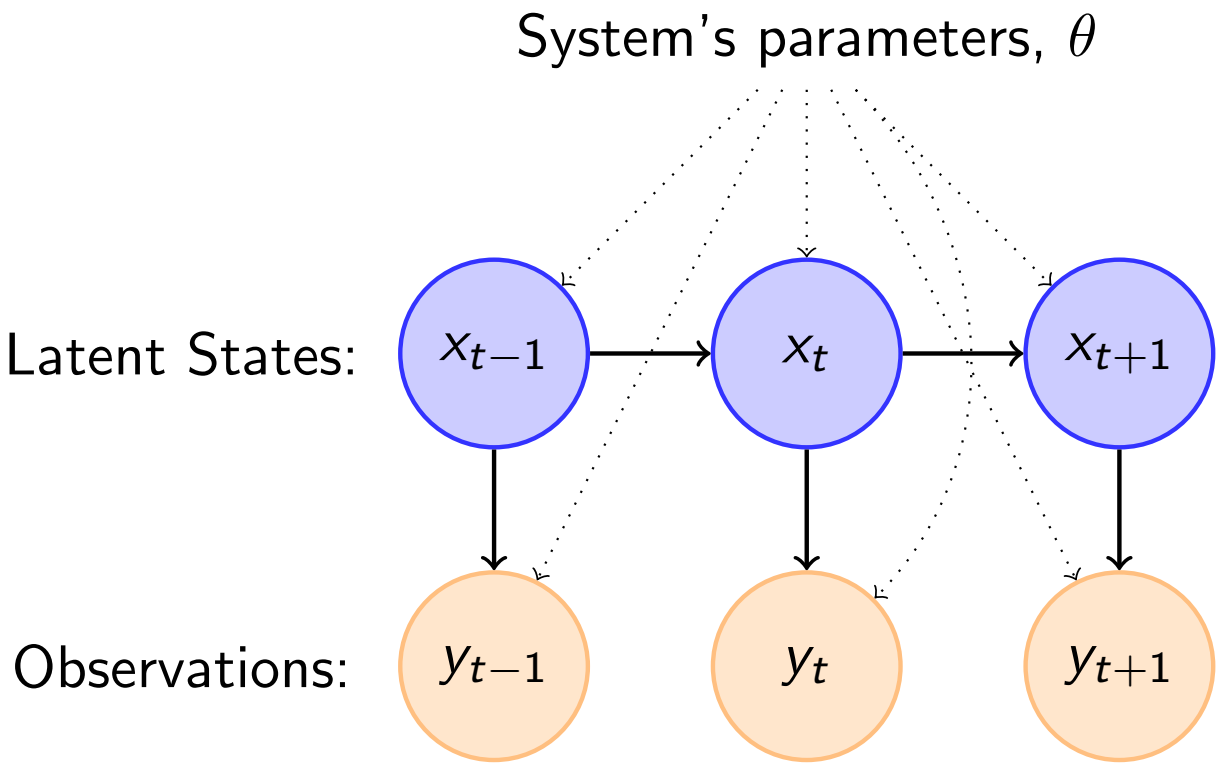
The Basic Story

1. The Bayesian analysis of covariance-matrix-valued state-space models can be difficult.
2. The subsequent model is computationally tractable, but it comes at a cost.

State-Space Models



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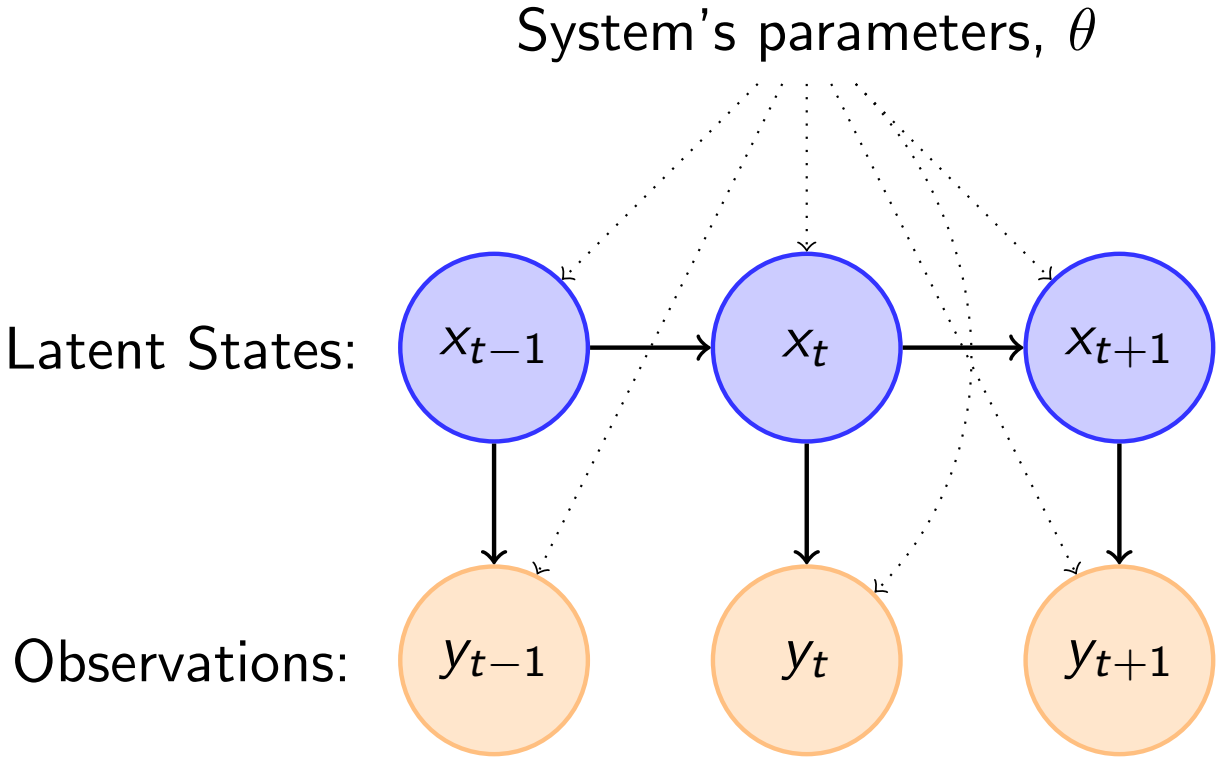
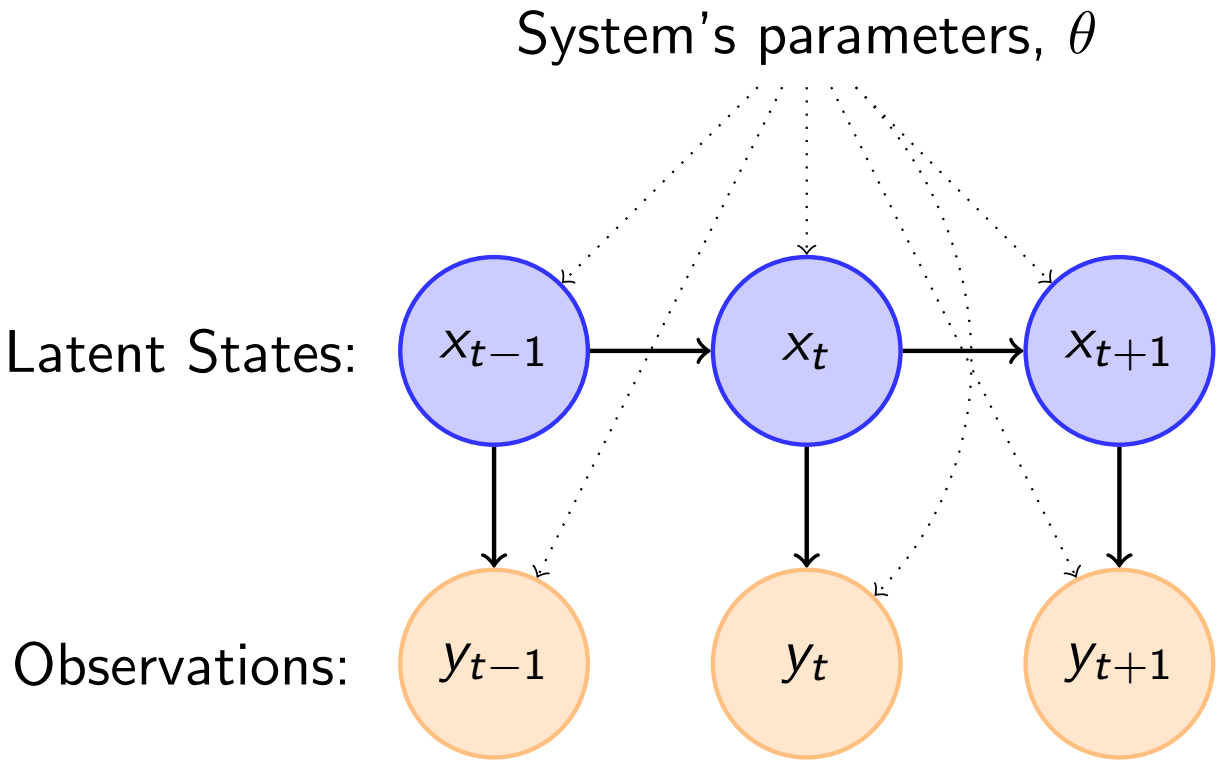


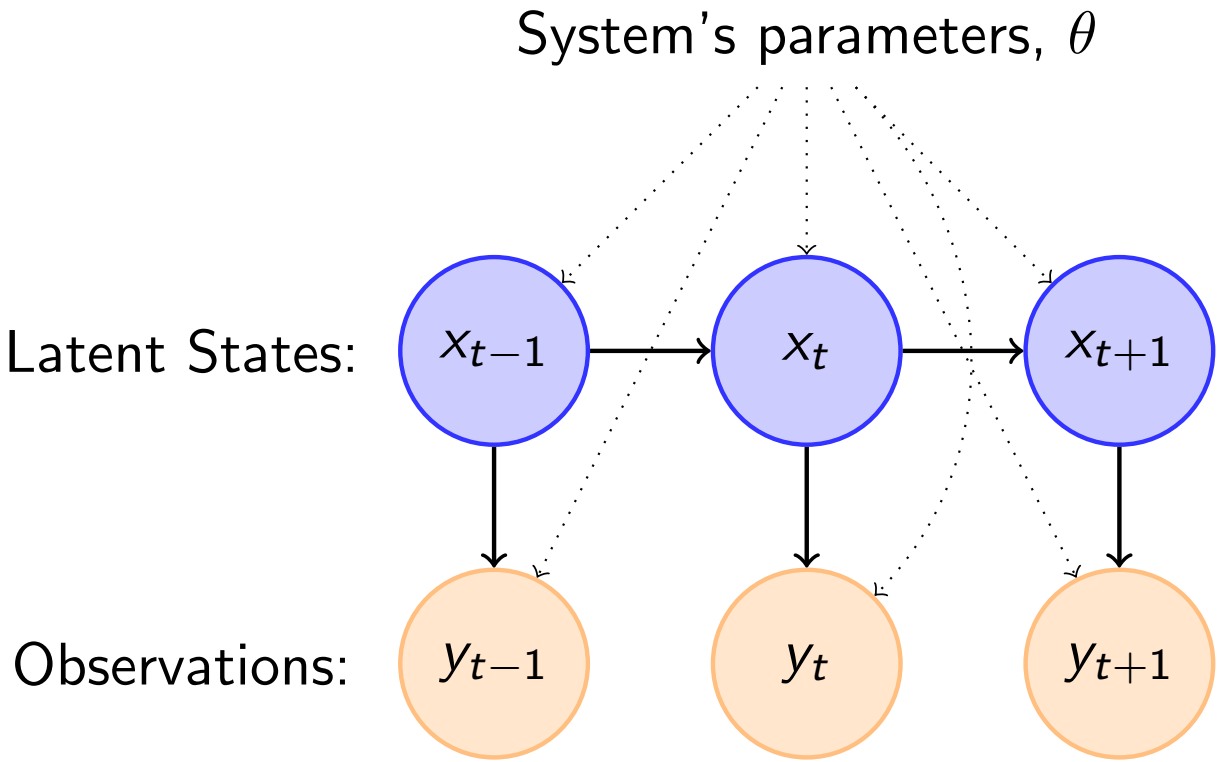
$$\left[\prod_{i=1}^{T} p(y_t|x_t,\theta)\right]\left[\prod_{i=2}^{T} p(x_t|x_{t-1},\theta)\right]p(x_1|\theta)$$

State-Space Models



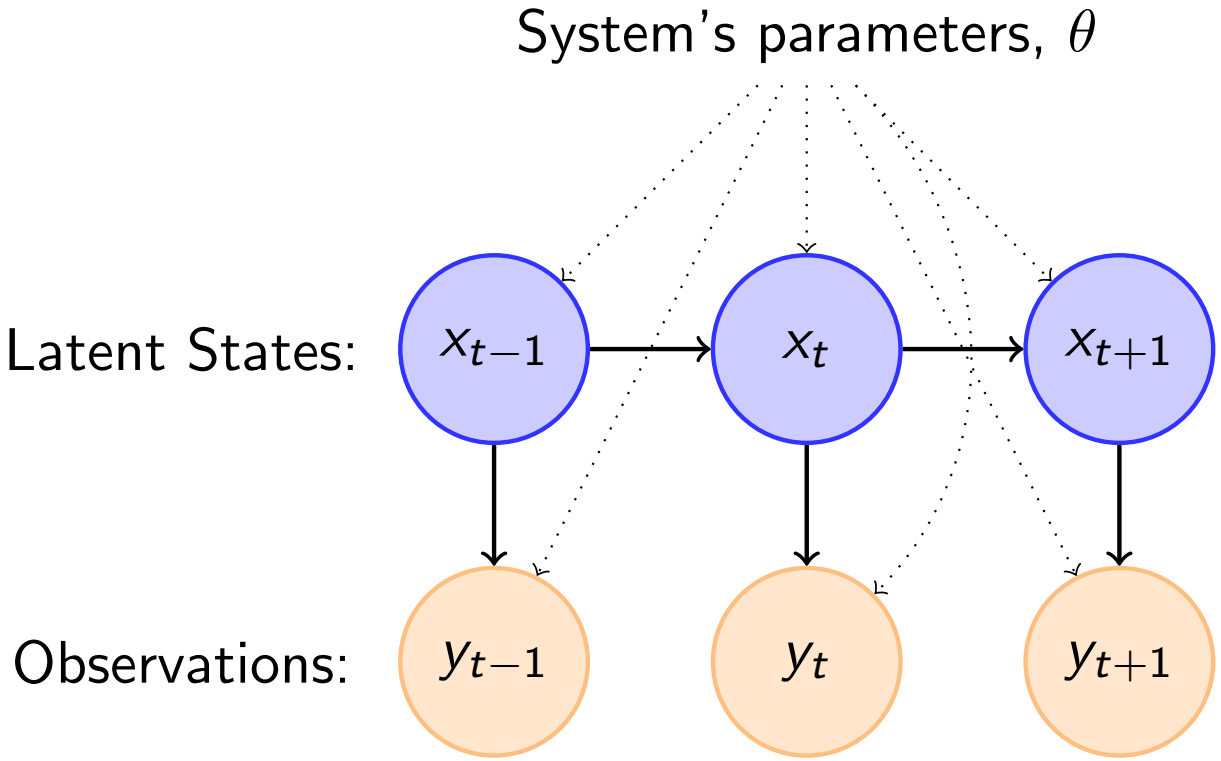
Filter: $p(x_t | y_{1:t})$.

State-Space Models



Smooth: $p(x_{1:T}|y_{1:T})$.

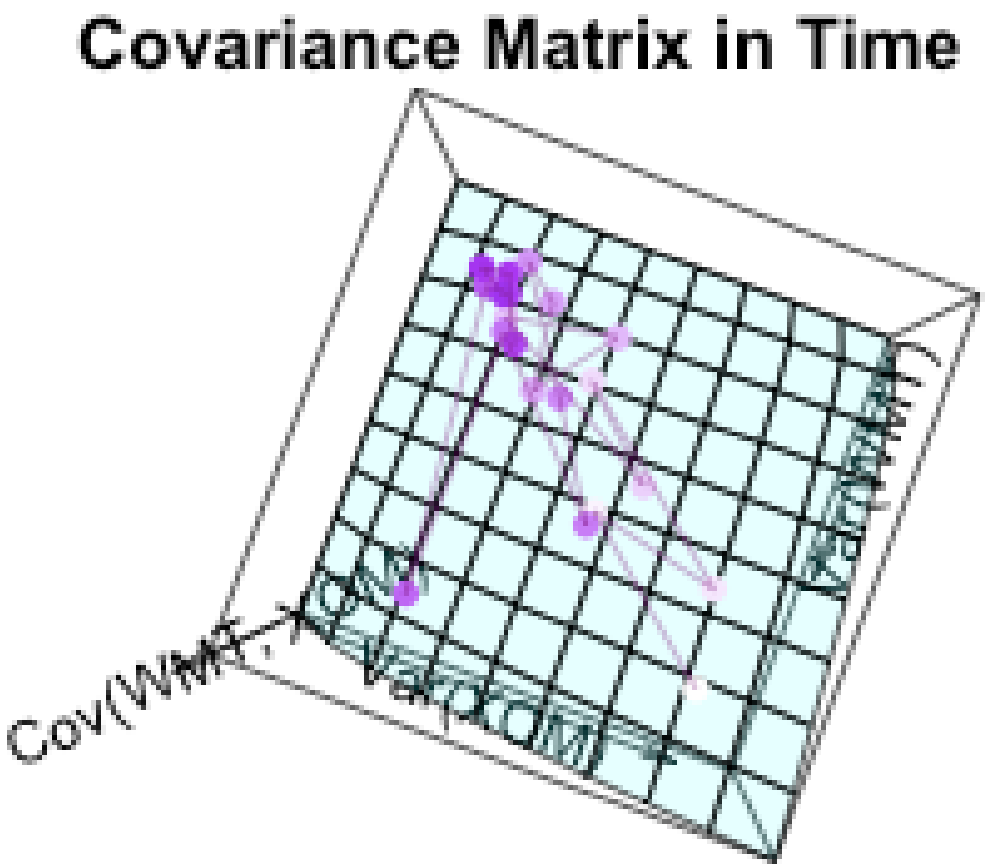
State-Space Models

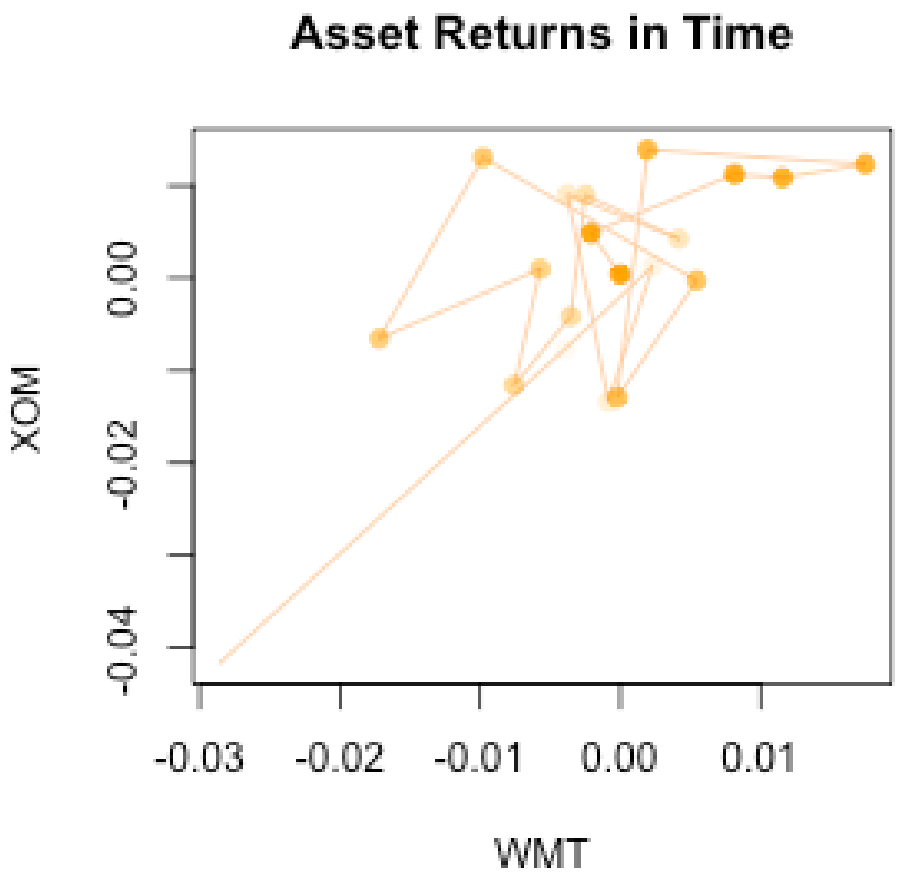


Infer: $p(\theta | y_{1:T})$.

State-Space Models in Finance

$$R_t \sim N(0, V_t),$$
$$V_t \sim \mathbb{P}(V_{t-1}).$$





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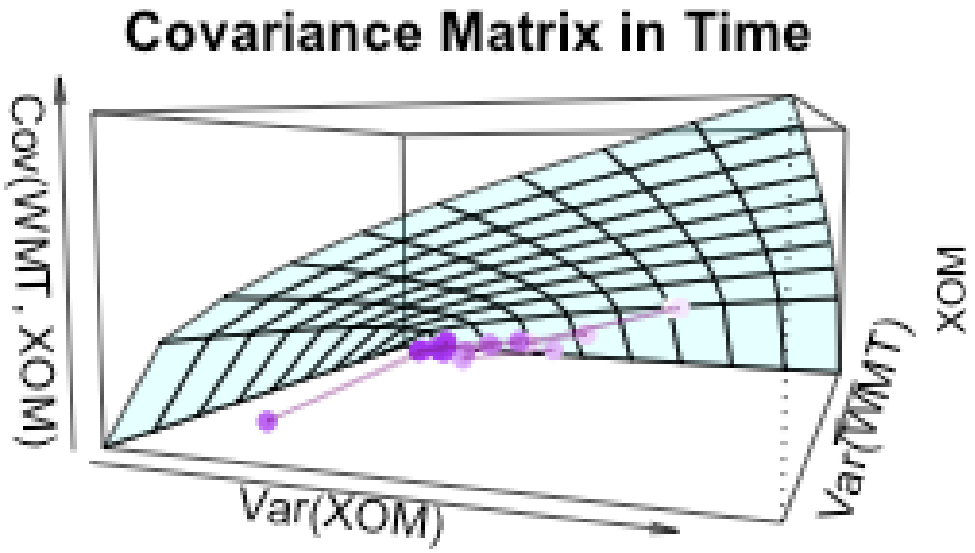

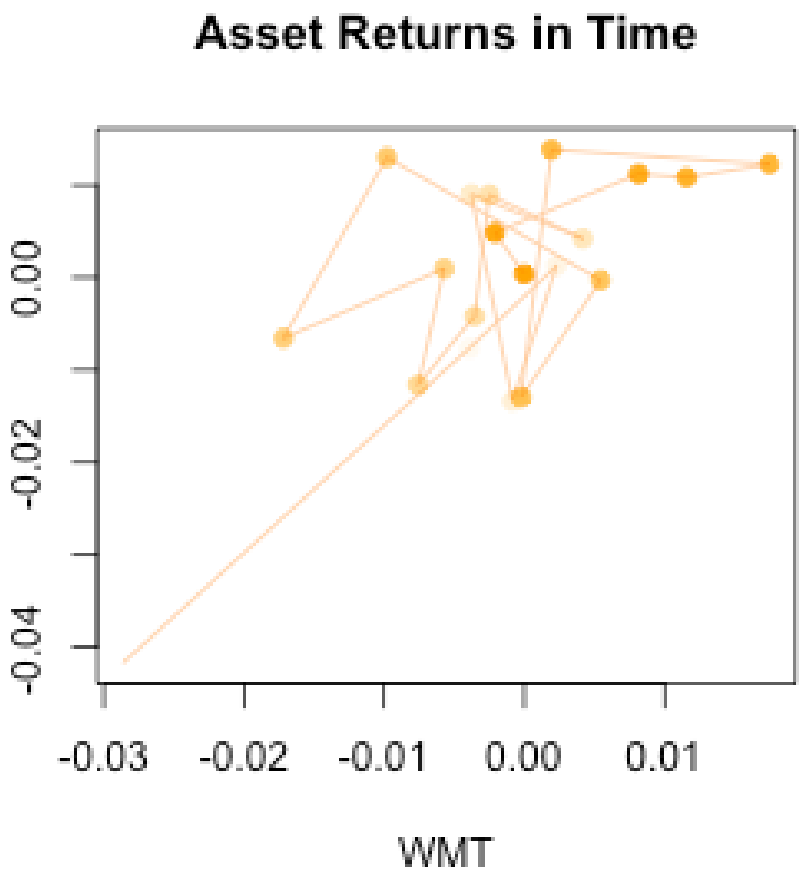

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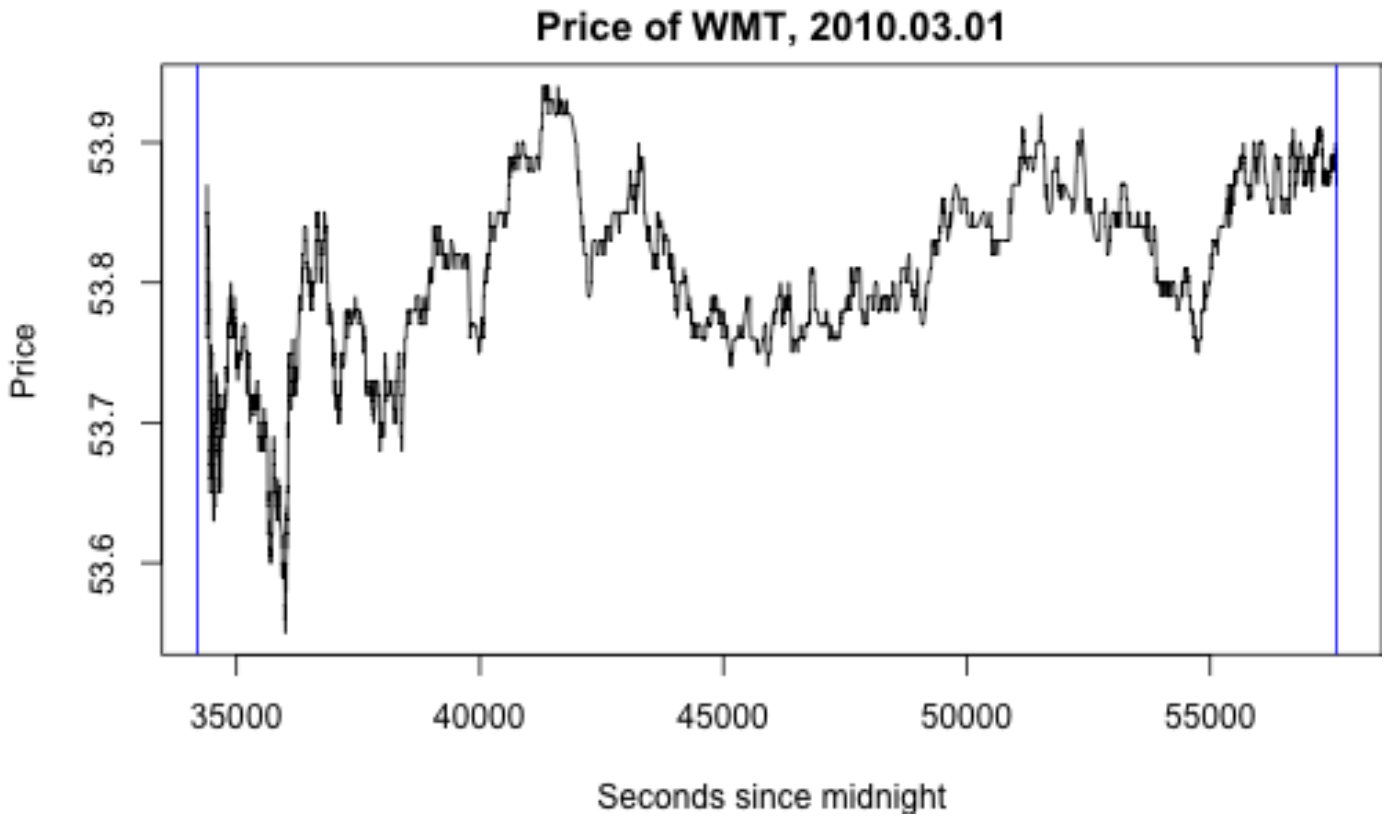

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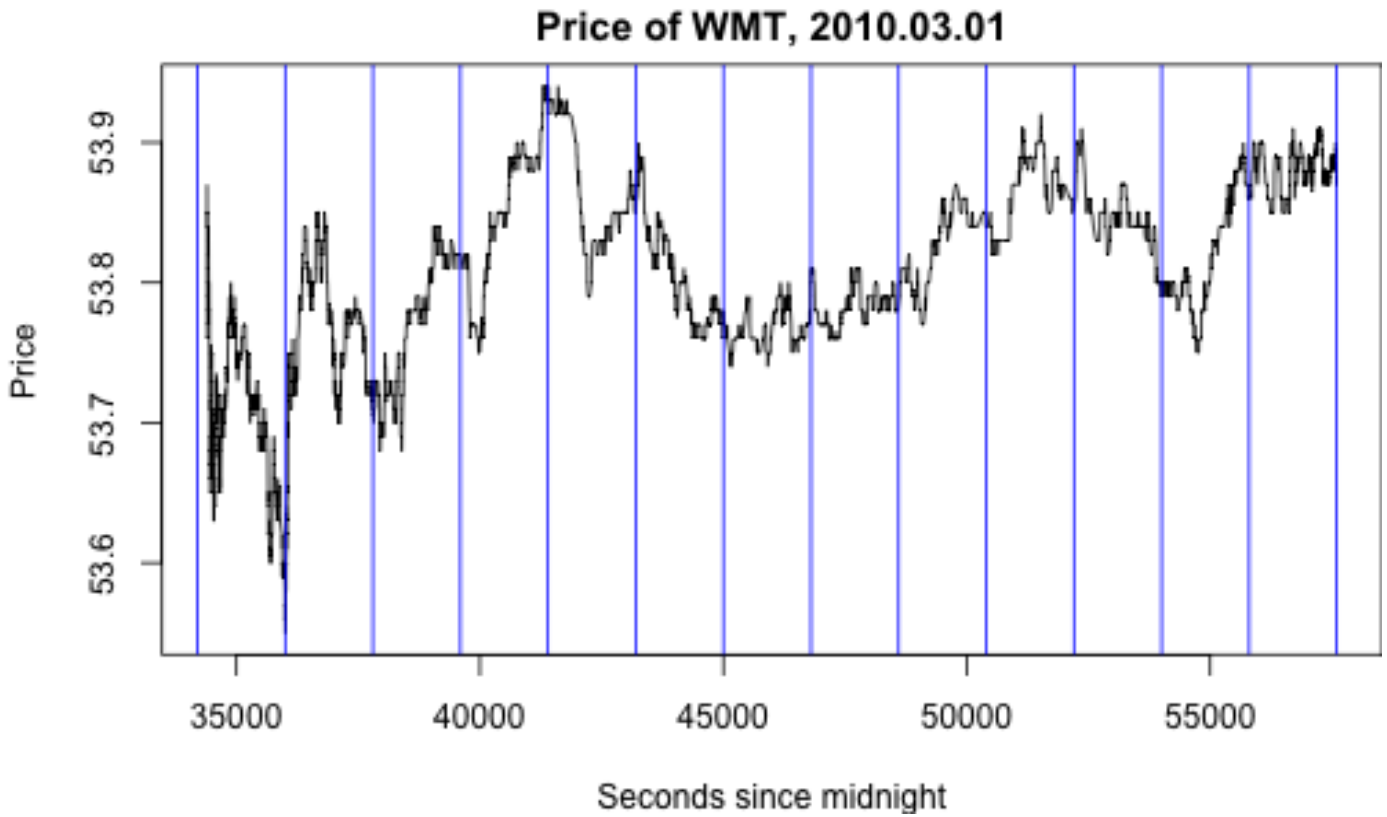
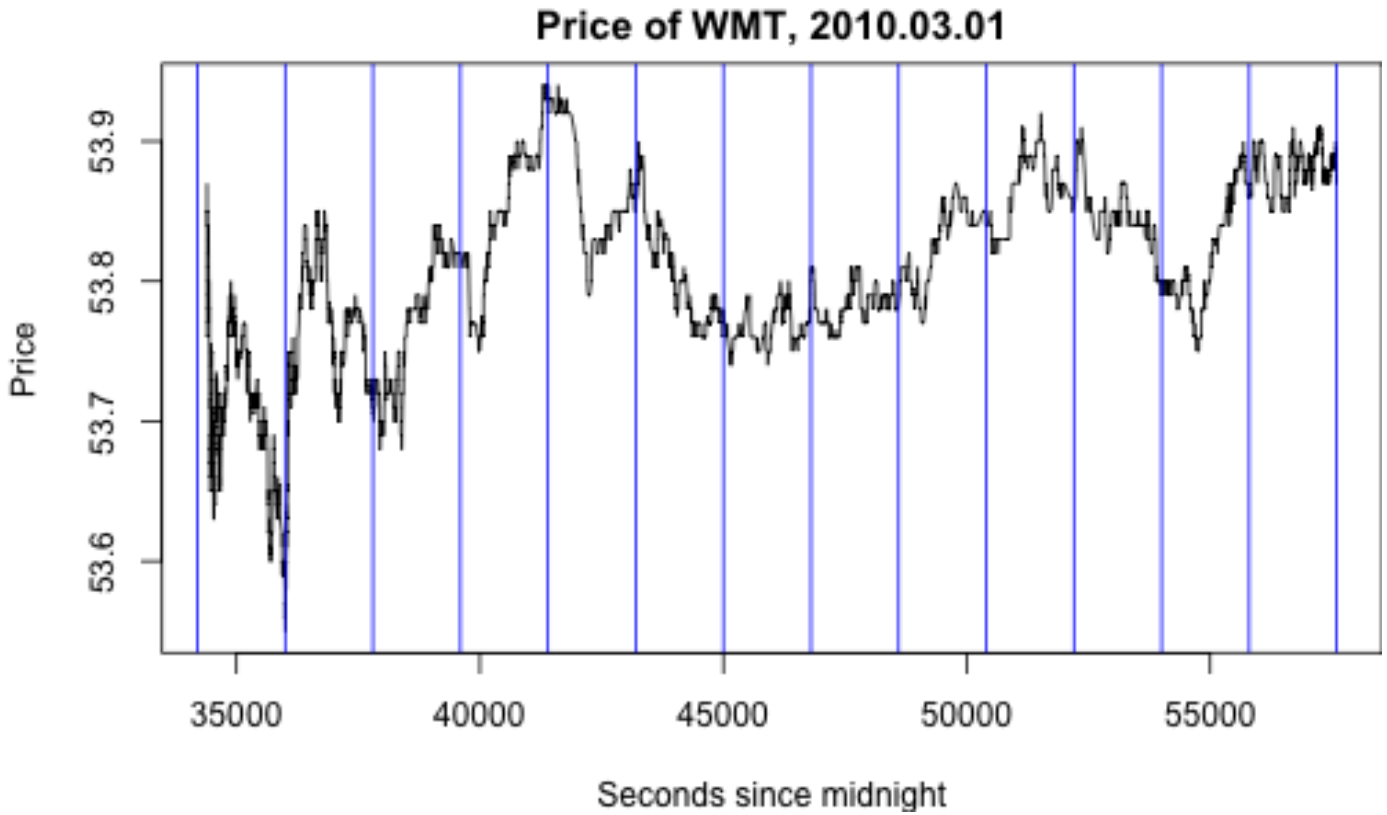
State-Space Models in Finance



$$R_{t,i} \sim N(0, V_t/k), \; i = 1, \ldots, k,$$
$$V_t \sim \mathbb{P}(V_{t-1}).$$

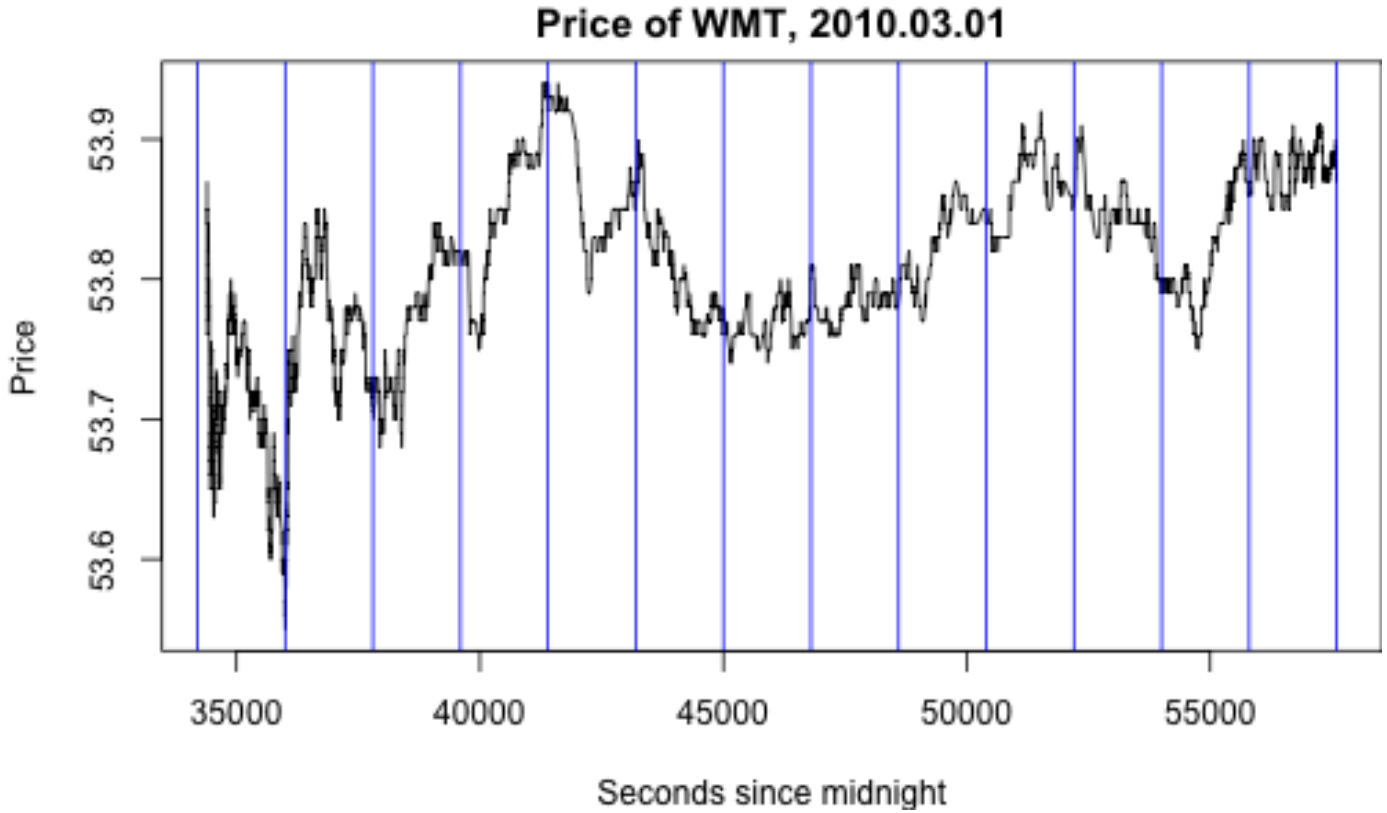
State-Space Models in Finance



$$Y_t \sim W_m(k, V_t/k), \;\; Y_t = \sum_{i=1}^{k} R_{t,i} R'_{t,i}$$

$$V_t \sim \mathbb{P}(V_{t-1}).$$

State-Space Models in Finance



$$Y_t \sim W_m(k, X_t^{-1}/k), \;\; Y_t = \sum_{i=1}^{k} R_{t,i} R'_{t,i}$$

$$X_t \sim \mathbb{P}(X_{t-1}).$$

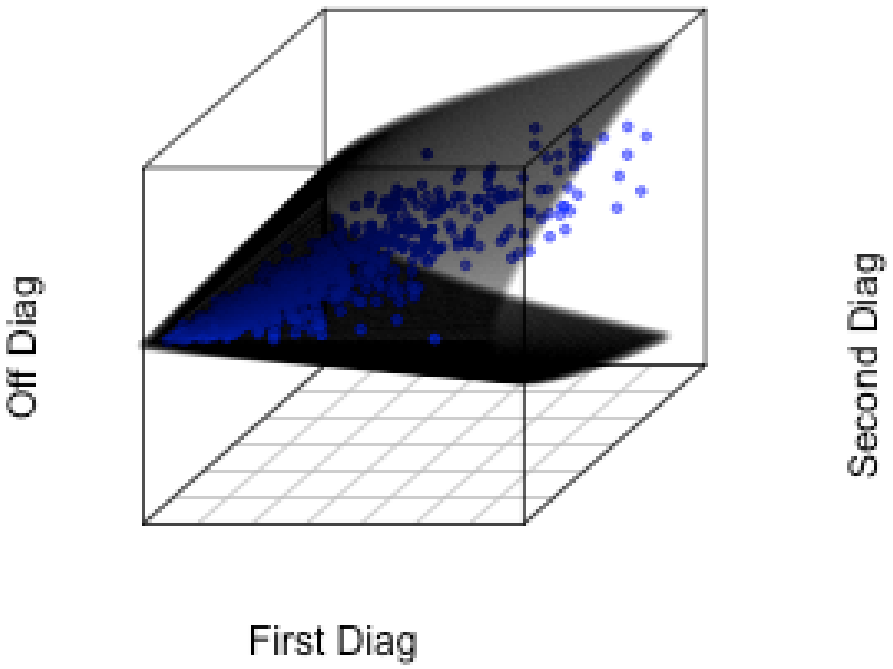
Our hands are now tied

$$\underbrace{\left[\prod_{i=1}^{T} p(Y_t|X_t,\theta)\right]}_{\text{Wishart}} \left[\prod_{i=2}^{T} p(X_t|X_{t-1},\theta)\right] p(X_1|\theta)$$

Problem: Moving around the state-space.

$$x_t = \text{Lower}(X_t) \sim \text{GP ?}$$

$$\begin{bmatrix} d_1 & c \\ c & d_2 \end{bmatrix}$$



Pick a new set of coordinates?

Matrix logarithm [Bauer and Vorkink, 2011]:

$$
\begin{aligned}
X_t &= U_t \exp(D_t) U_t', \\
\log X_t &= U_t D_t U_t', \\
Z_t &= \text{Lower}(\log X_t).
\end{aligned}
$$

$$
\underbrace{\Big[\prod_{i=1}^{T} p(Y_t|X_t,\theta)\Big]}_{\text{Wishart}} \Big[\prod_{i=1}^{T} p(X_t|X_{t-1},\theta)\Big] p(X_0|\theta)
$$

$$
p(X_{1:T}|Y_{1:T},\theta) \rightarrow \text{ Gibbs + Metropolis-Hastings.}
$$

$$
p(X_t|X_{-t},\, Y_{1:T},\theta).
$$

Pick a new set of coordinates?

LDL decomposition [Chiriac and Voev, 2010, Loddo et al., 2011]:

$$X_t = L_t \exp(D_t) L_t',$$
$$Z_t = (\ \text{StrictLower}(L_t), \text{Diag}(D_t)\ ).$$

$$\underbrace{\Big[\prod_{i=1}^{T} p(Y_t|X_t,\theta)\Big]}_{\text{Wishart}} \Big[\prod_{i=1}^{T} p(X_t|X_{t-1},\theta)\Big] p(X_0|\theta)$$

$$p(X_{1:T}|Y_{1:T},\theta) \rightarrow \text{ Gibbs + Metropolis-Hastings.}$$
$$p(X_t|X_{-t},\, Y_{1:T},\theta).$$

Use the original coordinates?

$$X_t = S_t \Psi_t S_t', \quad S_t S_t' = f(X_{t-1})$$

Source	$f(X_{t-1})$	Ψ_t	$p(X_{1:T} \mid Y_{1:T}, \theta)$
(1)	$\lambda^{-1} X_{t-1}$	$W_m(\rho, \mathbf{I}_m/\rho)$	
(2)	$\lambda^{-1} X_{t-1}$	$\beta_m\left(\frac{n}{2}, \frac{1}{2}\right)$	

(1) Philipov and Glickman [2006], Asai and McAleer [2009]
(2) Uhlig [1997], Rank m=1 Case Only

Other relevant work: Gourieroux et al. [2009], Fox and West [2011]; Prado and West [2010], Jin and Maheu [2013], Shirota et al. [2015]. GARCH literature... Bauwens et al. [2006].

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Uhlig Extension

$$X_t = S_t \Psi_t S_t', \quad S_t S_t' = \lambda^{-1} X_{t-1}$$
$$\Psi_t \sim \beta_m\left(\frac{n}{2}, \frac{k}{2}\right), \; k \in \mathbb{N};$$

Easy to compute:

- $p(X_t | Y_{1:t}, \theta)$ Wishart
- $p(X_t | Y_{1:t}, X_{t+1}, \theta)$ Shifted Wishart
- $p(X_{1:T} | Y_{1:T}, \theta)$

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Only need to record:

$$\Sigma_t = \lambda \Sigma_{t-1} + Y_t.$$

How does this work? Key Transformation

Muirhead [1982], Uhlig [1997], Díaz-García and Jáimez [1997]:

$$
\begin{tikzcd}
\text{Wishart} & & \text{Mult. Beta} \\
X_{t-1} \arrow[dr] & \perp & \Psi_t \arrow[dl] \\
& g \arrow[dl] \arrow[dr] & \\
\lambda X_t & \perp & Z_t \\
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Density of rank-deficient Wishart

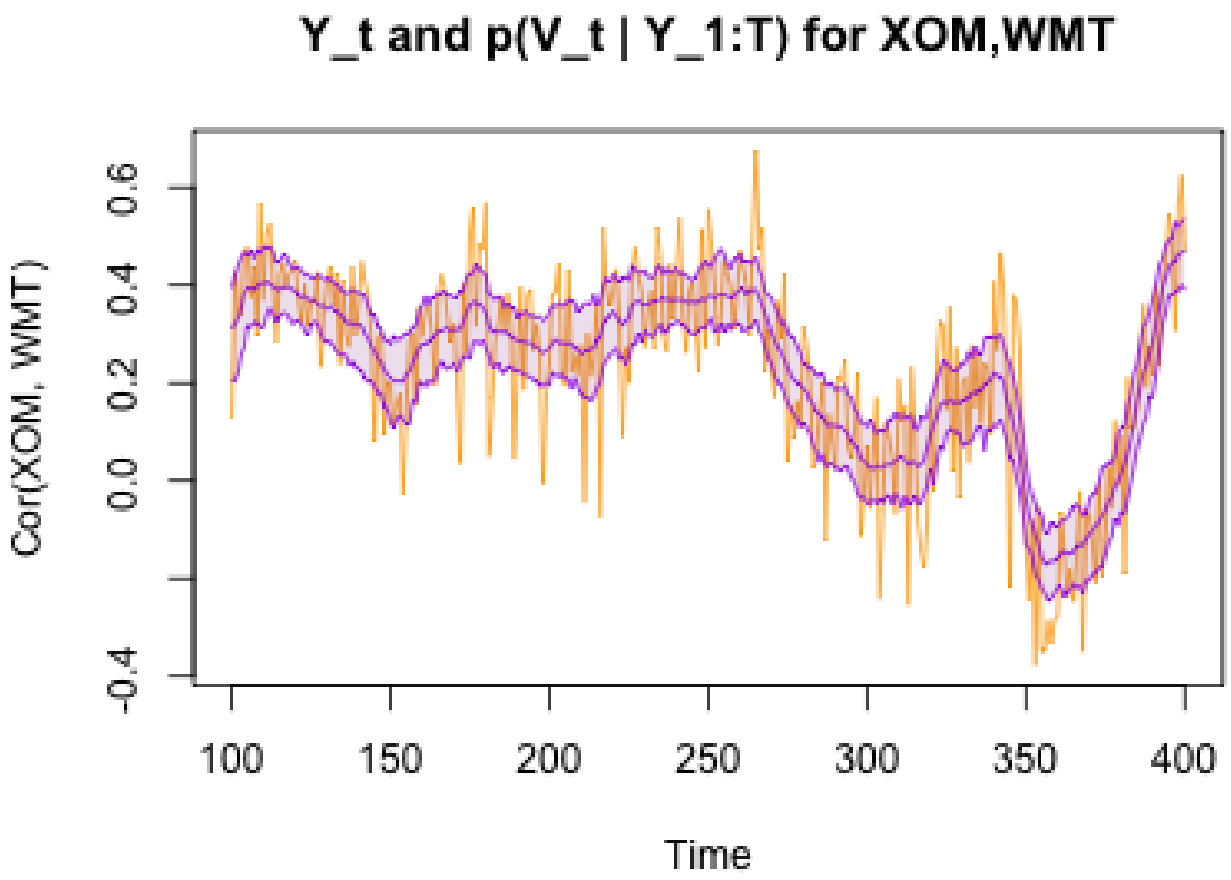
$$\frac{\pi^{-(mk-k^2)/2}|L|^{(k-m-1)/2}}{2^{mk/2}\Gamma_k\left(\frac{k}{2}\right)|V|^{k/2}} \exp\left(\operatorname{tr} \; -\frac{1}{2}V^{-1}Y\right)$$

$$(dY) = 2^{-k}\prod_{i=1}^{k} l_i^{m-k}\prod_{i<j}^{k}(l_i - l_j)(H_1' d\, H_1) \wedge \bigwedge_{i=1}^{k} dl_i.$$

(Introductory text: Mikusiński and Taylor [2002])

Example

- 30 stocks from DJIA as of Oct. 2010.
- Feb. 27, 2007 to Oct. 29, 2010.
- Y_t: Realized kernels (e.g. Barndorff-Nielsen et al. [2009])



Prediction Exercise

- Predictive portfolios:

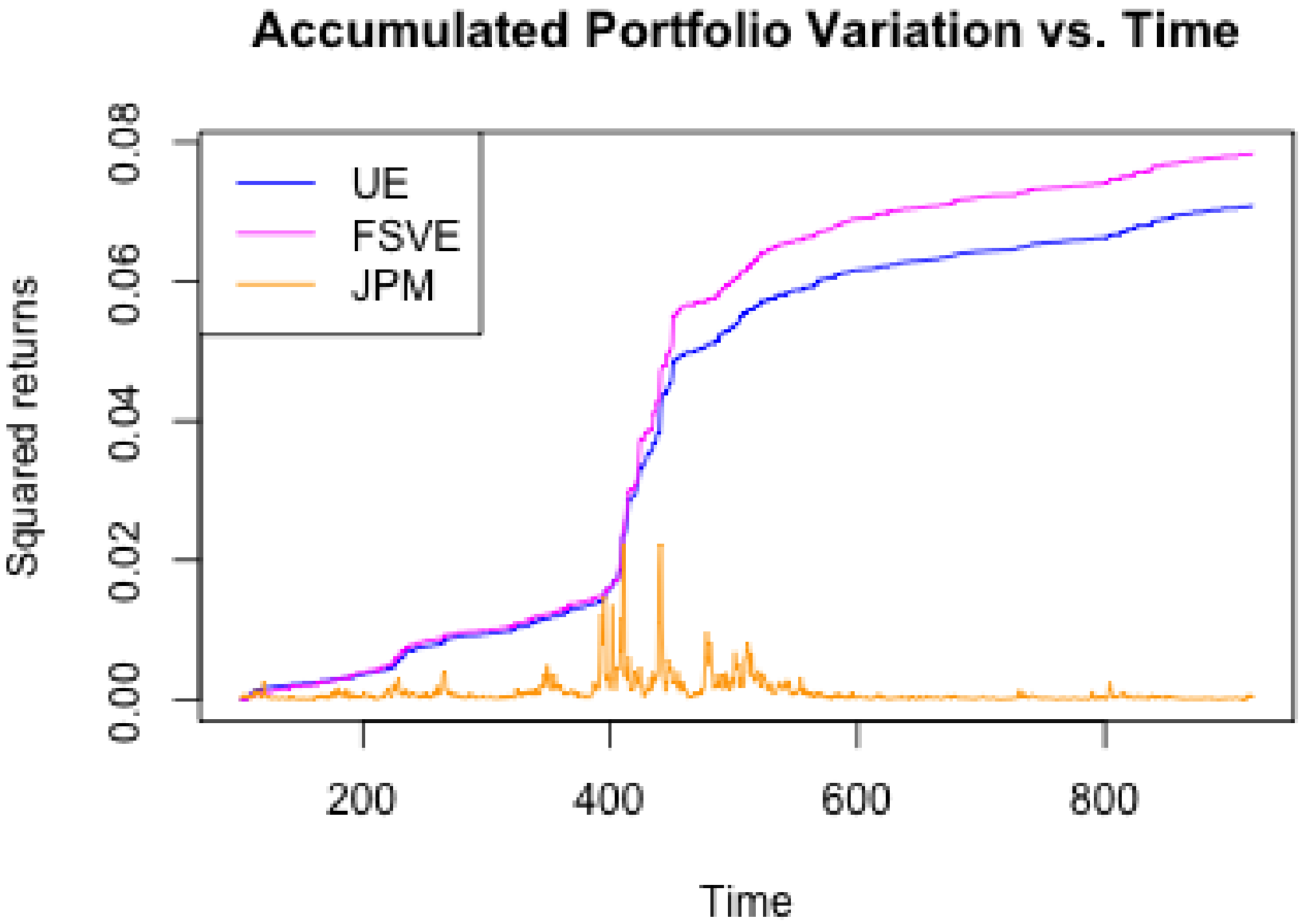
$$\pi_t^* = \underset{\pi'\mathbf{1}=1}{\text{argmin}}\ \pi' \hat{V}_t \pi$$
$$\hat{V}_t = \mathbb{E}[V_t | Y_{1:t-1}].$$

- Performance:

$$\text{portfolio variation} = \text{var}(\pi_t^{*\prime} r_t).$$

	root mean variation
FSV Extension	0.00977
Uhlig Extension	0.00936.

Prediction Exercise

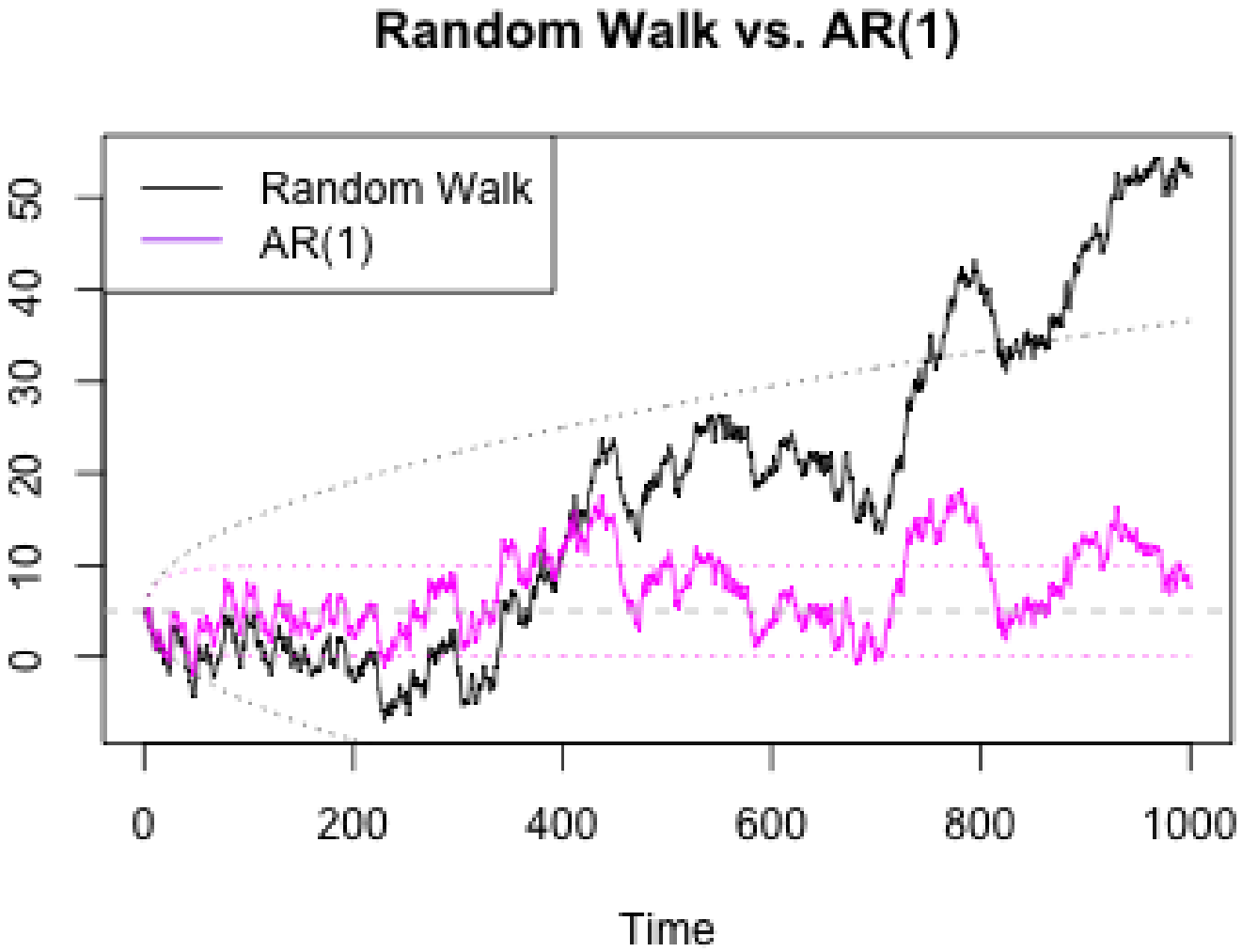


Drawbacks

Discussion:

- Roberto Casarin
- Catherine Scipione Forbes
- Enrique ter Horst, German Molina

Drawback: X_t is not stationary (realism)



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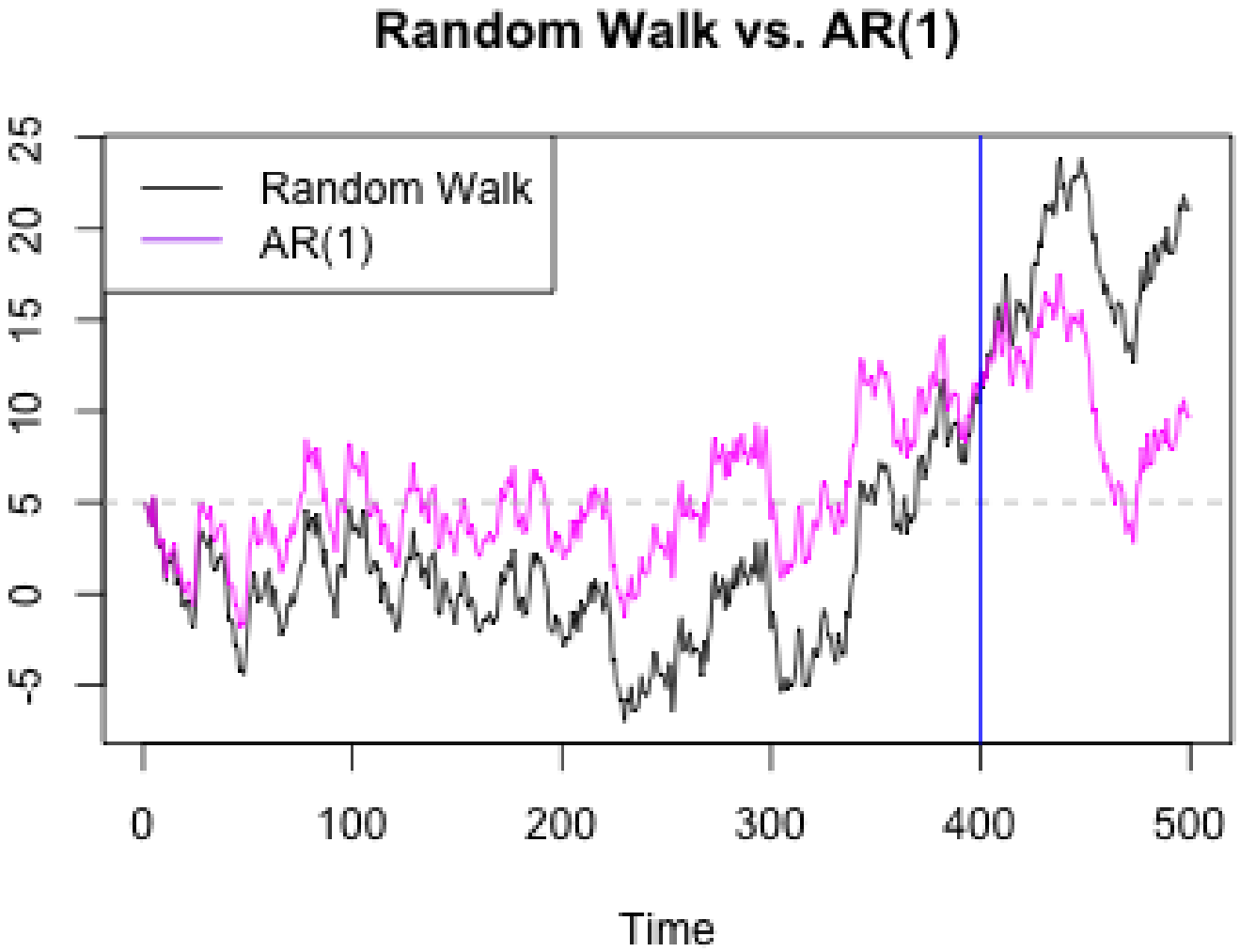

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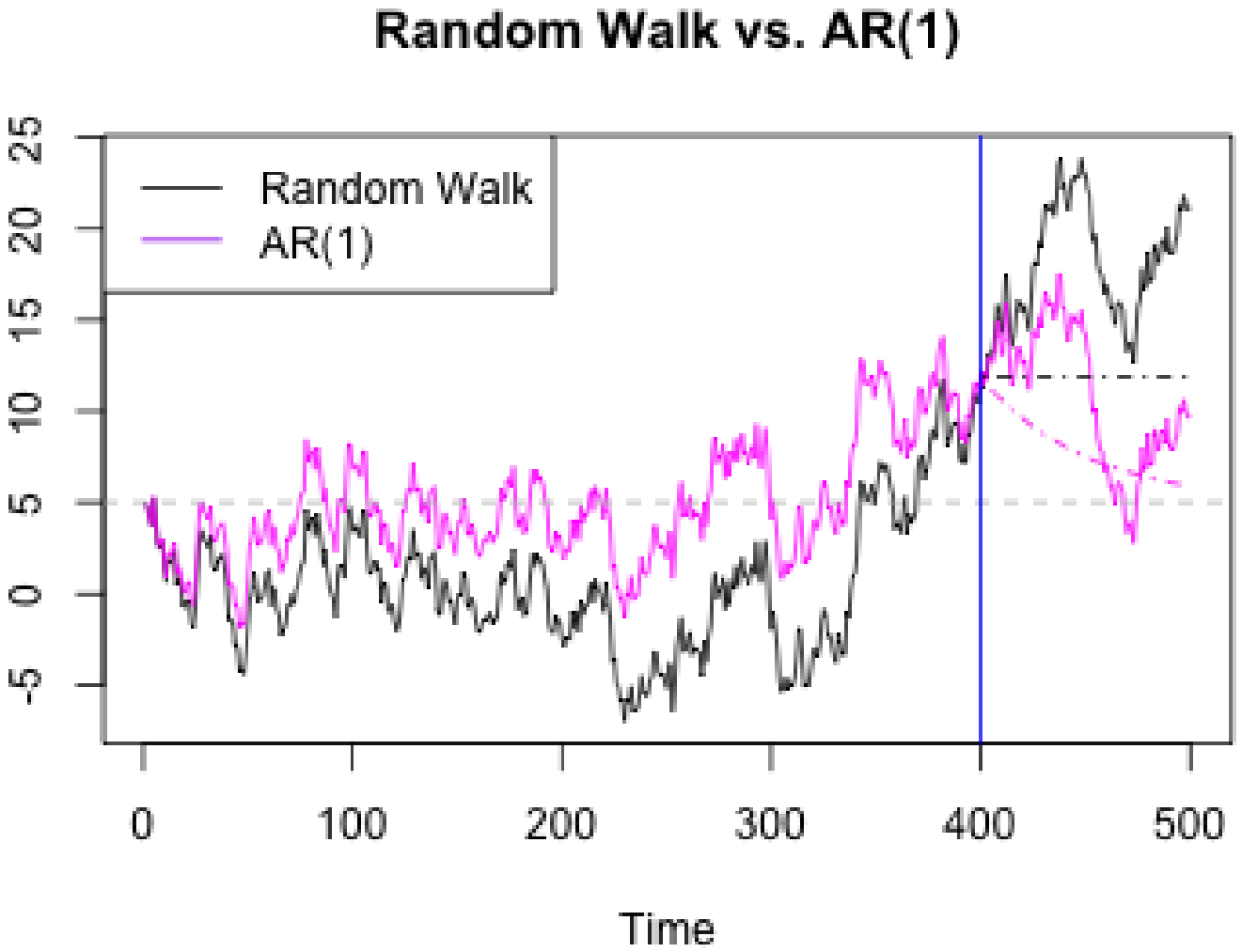


Drawback: X_t is not stationary (predictions)

Predictions of future variance:

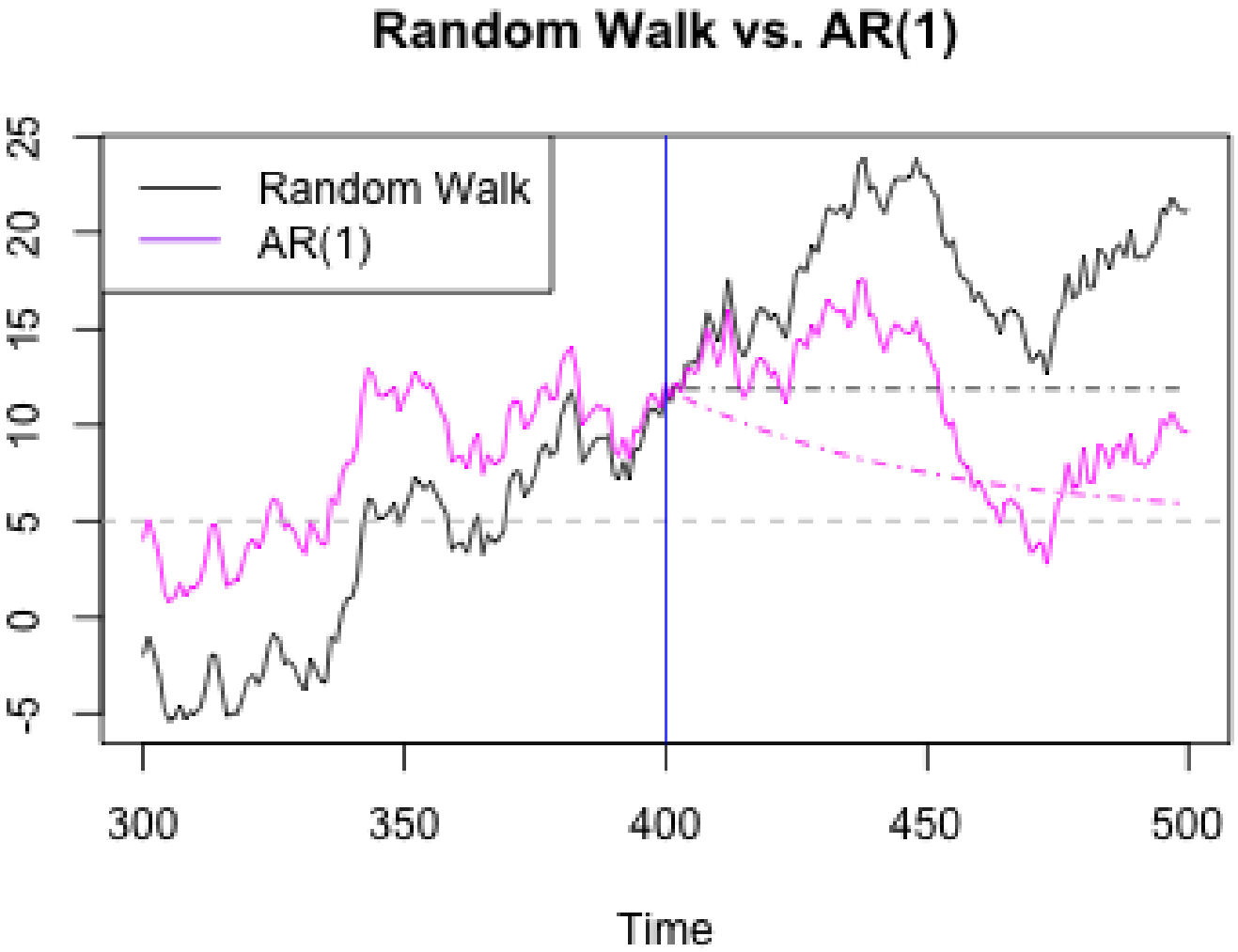
$$M_h = \mathbb{E}[X_{t+h}^{-1} \mid X_t^{-1}], \; h > 0.$$

Konno [1988]:

$$M_h = \frac{n+k-m-1}{n-m-1} \, \lambda \, M_{h-1}$$

where $M_0 = X_t^{-1}$.

What does this work at all?



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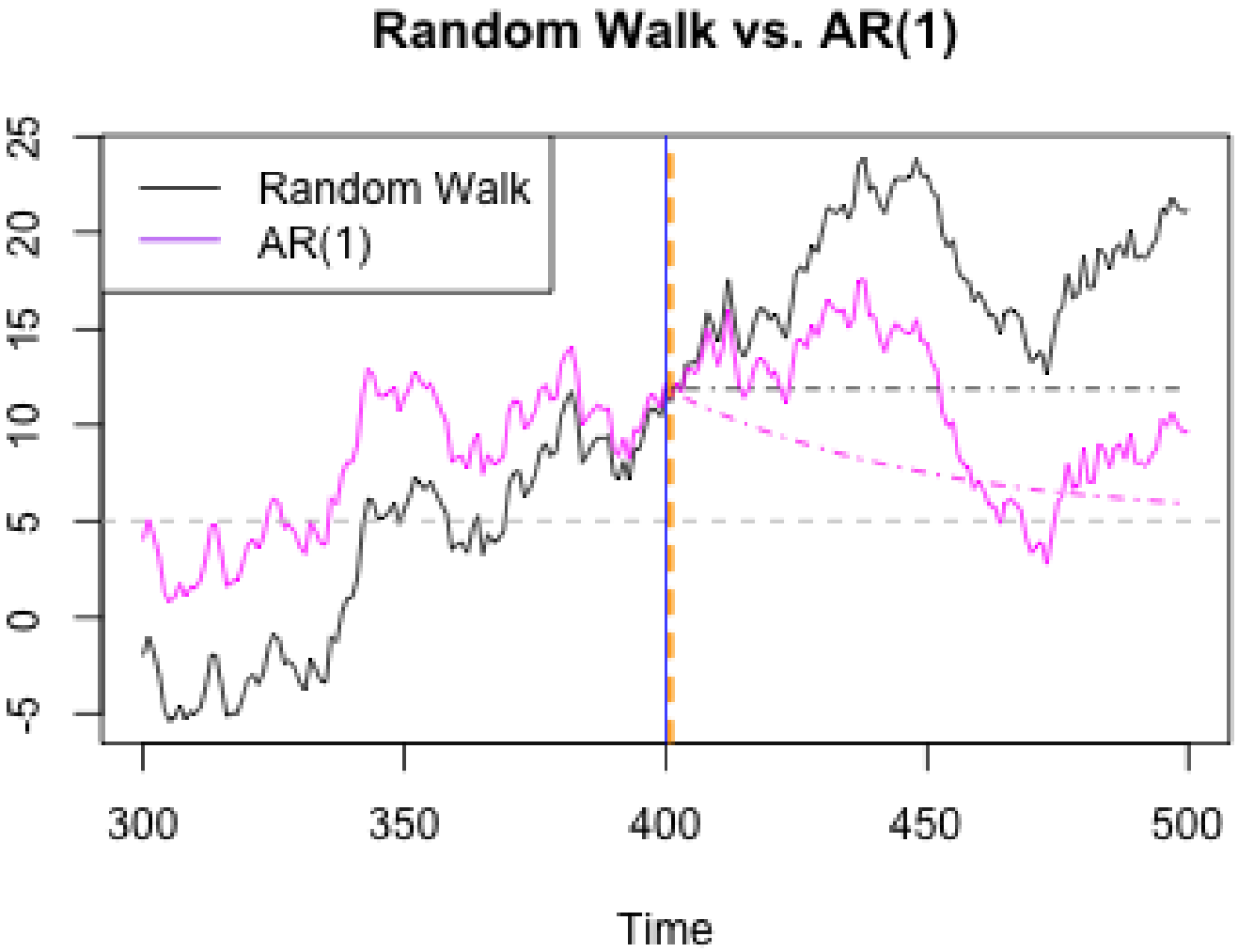

Volatility models: think in terms of forecasts

- Uhlig extension :

$$\mathbb{E}[X_{t+1}^{-1} | Y_{1:t}, \theta] = \frac{\lambda k}{n-m-1} \Big(\sum_{i=0}^{t-1} \lambda^i Y_{t-i} + \lambda^t \Sigma_0 \Big).$$

Volatility models: think in terms of forecasts

- Uhlig extension (EWMA):

$$\mathbb{E}[X_{t+1}^{-1}|Y_{1:t},\theta] = \frac{\lambda k}{n-m-1}\Big(\sum_{i=0}^{t-1}\lambda^i Y_{t-i} + \lambda^t \Sigma_0\Big).$$

$$\frac{n+k-m-1}{n-m-1}\,\lambda = 1 \implies$$

$$\mathbb{E}[X_{t+1}^{-1}|Y_{1:t},\theta] = (1-\lambda)\Big(\sum_{i=0}^{t-1}\lambda^i Y_{t-i} + \lambda^t \Sigma_0\Big).$$

Volatility models: think in terms of forecasts (continued)

- "GARCH" (EWMA-MR):

$$\mathbb{E}[X_{t+1}^{-1} | Y_{1:t}, \theta] \simeq (1-\gamma)C \; + \; \gamma\,(1-\lambda)\Big(\sum_{i=0}^{t} \lambda^i Y_{t-i}\Big).$$

- Univariate stochastic volatility:

EWMA-MR of the log squared returns

- Leverage effects:

asymmetrically weight past observations
depending on market movements.

Estimating $\theta = (n, k, \lambda, \Sigma_0)$

The model:

$$
\begin{aligned}
Y_t &= W_m(k, (kX_t)^{-1}), \\
X_t &= S_t \Psi_t S_t', \quad S_t S_t' = \lambda^{-1} X_{t-1}, \\
\Psi_t &\sim \beta_m\Big(\frac{n}{2}, \frac{k}{2}\Big), \; k \in \mathbb{N}.
\end{aligned}
$$

Conjugate prior:

$$X_1 \sim W_m(n, (\lambda\, k\, \Sigma_0)^{-1}).$$

$$\Sigma_t = \sum_{i=0}^{t-1} \lambda^i Y_{t-i} + \lambda^t \Sigma_0$$

$$Y_{-\tau}, \ldots, Y_0, Y_1, \ldots, Y_T.$$

Estimating $\theta = (n, k, \lambda, \Sigma_0)$

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Conjugate prior:

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$$\Sigma_t = \sum_{i=0}^{t-1} \lambda^i Y_{t-i} + \lambda^t \Sigma_0 \;\rightarrow\; \Sigma_0(\lambda) = \sum_{i=0}^{-\tau} \lambda^i Y_{-i} + 0.$$

$$Y_{-\tau}, \ldots, Y_0, Y_1, \ldots, Y_T.$$

Recapitulation

1. Given our specific observation distribution, it isn't easy to construct tractable matrix-valued state-space models.
2. Uhlig essentially provides a way to do this, but it comes with a cost.

Slides with references:

`http://www.jessewindle.com/`

Thank you for your attention.

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